

Project Proposal (Proposed Solution) Report

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	5 Marks

Executive Summary & Solution Objectives:

AI FitTrack API is an intelligent, production-ready backend application built with Node.js, Express.js, MongoDB Atlas (Mongoose ODM), JWT Authentication, bcrypt, and the Google Gemini AI SDK (@google/genai v2.x). It enables users to register securely, log daily workout routines across 6 distinct categories, search past activities using keyword and date regex filters, and receive customized AI workout routines and fitness progress insights directly from Google Gemini AI.

Core System Features & Functional Modules:

Feature Module	Operational Purpose	Technical Implementation	Developer Benefit
Authentication & Security	User registration, password encryption, and JWT token issuance.	bcryptjs (10 salt rounds), jsonwebtoken, auth.js middleware guard.	Stateless security and strict user data isolation.
Workout Management	Log, retrieve, update, and delete daily fitness activities.	Mongoose Workout schema, user ObjectId referencing, date sorting.	Structured tracking across Cardio, Strength, Yoga, Running.
Multi-Criteria Search	Instant query by workout name, category, or date format.	Case-insensitive MongoDB regex queries in workoutController.js.	High usability for fitness dashboards and mobile apps.
AI Workout Plans	Dynamic personalized exercise routines tailored to user goals.	Google Gemini 1.5 Flash SDK prompt synthesis in geminiService.js.	Eliminates need for expensive personal trainer software.
AI Fitness Insights	Intelligent analysis of workout consistency and calorie burn.	Aggregated metric prompt evaluation via Gemini AI engine.	Actionable progress tracking that motivates users.

Proposed Solution (Continued) — Resource Requirements

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Resource Requirements & Environment Specifications:

Resource Type	Component / Tool	Specification / Allocation	Purpose in Solution
Hardware	Development Workstation	x86/ARM CPU, 8GB+ RAM, High-speed Internet	Local development, testing, and Git version control.
Software	Node.js Runtime	v20.x or higher LTS	Asynchronous JavaScript execution environment.
Software	Express Web Framework	v4.19.2	HTTP routing, middleware pipeline, and JSON parsing.
Software	Testing & Verification	Postman Collection v2.1 & Thunder Client	Automated API request evaluation and response testing.
Cloud DB	MongoDB Atlas Cluster	M0 Free Shared Tier (TLS 1.3)	Persistent cloud storage for users and workout data.
Cloud AI	Google AI Studio	Gemini 1.5 Flash API Key	Generative AI LLM inference for fitness recommendations.

Risk Assessment & Mitigation Matrix:

Risk Category	Potential System Impact	Mitigation Strategy Implemented in AI FitTrack API
AI Rate Limits	Throttling during traffic spikes on Gemini API.	Concise prompt design, efficient token usage, and graceful fallback handling.
Data Tampering	Users viewing or editing other users' workouts.	Strict user ownership validation in workoutController (req.user.id == workout.user).
Database Latency	Slow response times on large workout history sets.	Indexed fields on user and workoutDate; connection reuse in db.js.